Examiner only

Answer **all** questions.

1. (a) A group of students was asked to find the density of an object.
They made the following measurements.

mass of the object = 15 g
volume of water in measuring cylinder = 20 cm^3
volume of water and object = 25 cm^3

Use the results to answer the following questions.

(i) Calculate the volume of the object. [1]

volume = ... cm^3

(ii) Use an equation from page 2 to calculate the density of the object. [2]

density = ... g/cm^3

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(iii) The teacher said the object was made from one of the materials in the table below.

Material	Density (g/cm^3)
cork	0.2
magnesium	1.7
aluminium	2.7
steel	7.8
iron	7.8
copper	8.5
lead	11.3

Ffion thought the object was made of aluminium.
Use the data in the table to explain whether you agree with Ffion. [2]

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(b) The group is given a solid object that floats on water.
Describe how they would change the experiment to find the density of this object.
They still use the measuring cylinder partly filled with water. [3]

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